

1. Project Identity

...

Project Name:

We Eat

Type:

Food sharing marketplace application

Purpose:

Connect people and organizations to share,
sell, exchange and rescue surplus food.

...

2. Core Business Logic

Explain:

Three Listing Types

FREE

...

Provider gives food without payment.

'''

Example:

'''

Homemade food surplus

Restaurant donation

Community sharing

'''

DISCOUNTED

'''

Provider sells surplus food below original price.

'''

Example:

'''

Original price:

500 BDT

Pay:

200 BDT

...

EXCHANGE

...

Users exchange food items.

...

Example:

...

Cake

Exchange request:

Vegetables

...

3. User System

Explain:

Database Roles

...

ADMIN

MODERATOR

USER

...

Anonymous users:

...

No database role.

No authentication session.

...

Explain permission matrix:

Feature	Anonymous	User	Moderator	Admin
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- - - -

Browse food	✓	✓	✓	✓
-------------	---	---	---	---

Post food	X	✓	✓	✓
-----------	---	---	---	---

Delete listing	X	Own only	✓	✓
----------------	---	----------	---	---

Manage users	X	X	Limited	✓
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4. Complete Architecture

Explain:

...

Mobile Browser

|

|

Next.js Frontend

|

|

Next.js API Proxy

|

|

FastAPI Backend

|

|

Neon PostgreSQL

|

|

Cloudinary Storage

...

5. Frontend Architecture

Explain:

Technology:

...

Next.js

TypeScript

App Router

React Components

CSS Modules

Framer Motion

...

Explain folder meaning:

...

app/

├─ public pages

├─ authentication pages

├─ dashboard pages

├─ admin pages

└─ API proxy

...

6. Frontend → Backend Communication

Explain exact flow:

Example:

User clicks login:

...

Login Form

↓

Next.js server route

/api/auth/login

↓

FastAPI

/api/v1/auth/login



Database verification



JWT generated



HTTP-only cookie



User authenticated

...

7. Backend Architecture

Explain:

...

FastAPI

```
|
|
├── API Routes
|
|
├── Authentication
|
|
├── Database Layer
|
|
├── Schemas
|
|
└── Services
...
---
```

8. FastAPI Working Method

Explain:

Request lifecycle

Example:

```
...
```

HTTP Request

↓

Router



Authentication Middleware



Schema Validation



Business Logic



Database Query



Response Schema



JSON Response

...

9. Database Connection

Explain:

...

FastAPI

↓

SQLAlchemy Async Engine

↓

asyncpg driver

↓

Neon PostgreSQL

...

Include:

Connection pooling

Environment variables

Session management

Transactions

10. Authentication System

Detailed:

Registration

...

User enters data

↓

OTP generated

↓

Email service sends OTP



OTP verified



Password hashed



User created



JWT issued

...

Login

...

Email/password



Verify hash



Create JWT



Store in HTTP-only cookie



Session available

...

11. JWT Structure

Explain:

Payload:

```
```json
{
 "user_id":123,
 "role":"USER",
 "exp":"time"
}
```
```

Rules:

Never trust role only from token.

Always check current database status.

12. Food Listing Flow

Explain:

Create listing:

...

User uploads image



Cloudinary upload



Receive URL



Save listing in Neon



Listing appears

...

13. Cloudinary System

Explain:

Purpose:

...

Database stores:

image URL

Cloudinary stores:

actual file

...

Folder strategy:

...

we-eat/

|

users/

|

listings/

|

reviews/

...

14. Comment System

Explain:

Database:

'''

comments

id

listing_id

user_id

parent_comment_id

content

is_deleted

'''

Tree building:

'''

Flat database rows

↓

Frontend converts

↓

Nested replies

...

15. Review System

Explain:

Important rule:

...

Only completed transactions can create reviews

...

Database:

...

review

order_id

reviewer

reviewee

rating

comment

...

16. Favorite System

Explain:

Database only.

Never use localStorage as source of truth.

Flow:

...

Click save



API request



Database insert/delete



UI update

...

17. Order System

Explain:

Flow:

...

Request food



Provider accepts



Pickup/delivery



Complete



Review unlocked

...

18. Exchange System

Explain:

...

Listing



Exchange request



Accept/reject



Complete exchange



Review

...

19. Admin System

Explain:

Admin controls:

...

Users

Listings

Reports

Moderators

Analytics

...

20. Moderator System

Explain:

...

Reports

↓

Investigation

↓

Action

↓

Audit log

...

21. Coding Rules for AI Agents

Very important section.

Example:

...

Before adding a new feature:

1. Check existing architecture
2. Reuse existing patterns
3. Do not create unnecessary files
4. Keep API naming consistent
5. Validate permissions
6. Never bypass database checks

...

22. Security Rules

Explain:

Never:

...

Store JWT in localStorage

Expose private addresses

Trust frontend validation

Upload files without checks

Allow direct database access

...

23. Development Philosophy

Explain:

The project prefers:

...

Simple architecture

Clear separation

Reusable components

Strong database constraints

Secure authentication

Mobile-first design

...

24. AI Agent Instructions

Final section:

...

When modifying We Eat:

First understand:

1. Database schema
2. API flow
3. Authentication system
4. Existing components

Do not rewrite architecture.

Do not introduce unnecessary dependencies.

Follow existing patterns.

...